

Patrick DENG

DATA ANALYSIS AND MLOPS ENGINEER

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Professional Experience

MLOps & Data Engineer at Walmart Canada | Mississauga ON 2024.07 - Present

- Spearheaded efforts to link web and in-store customer experiences by designing **scalable data pipelines** in **GCP BigQuery**, leveraging **data modeling**, **ETL best practices**, and **SQL** to enhance cross-channel marketing strategies and improve customer engagement.
- Automated **data transformation workflows** using **BigQuery SQL**, **Python**, and **Airflow**, ensuring scalability and efficiency for processing **70M+ daily records**, reducing operational time by 30%.
- Partnered with cross-functional teams to conduct **deep-dive analyses** of large-scale customer data, delivering actionable insights through interactive dashboards and reports, driving data-driven business decision-making.
- Explored and integrated **LLMs (GPT, Gemini APIs)** into data-driven applications, enabling advanced **NLP automation** and demonstrating a forward-thinking approach to leveraging AI for pipeline improvements.
- Improved data quality by collaborating with teams to optimize **back-end data sources**, ensuring accuracy, reliability, and simplicity in reporting workflows.
- Gained hands-on experience with **cloud platforms** and modern data engineering tools, with transferable skills.

Engineer & Analyst, Data, Strategy at Dane Creek Capital Corp. | Toronto ON 2021.07 - 2024.07

- Designed and implemented a **cloud-based data warehouse** using **Airflow (GCP)** to integrate diverse sources (CRM, ERP, analytics), enabling **scalable ETL pipelines** and improving reporting accuracy by 25%.
- Developed and maintained **interactive dashboards** using **Tableau**, **Looker Studio**, and **Metabase**, providing real-time visibility into key performance metrics and actionable insights for stakeholders.
- Built **predictive models** for customer behavior analysis (e.g., churn prediction, product recommendations), driving **business growth** by improving customer retention and optimizing product offerings.
- Optimized **data mining workflows** and financial forecasting processes by leveraging **Python**, **SQL**, and advanced analytical tools, reducing manual effort and improving operational efficiency.
- Championed the adoption of **cloud technologies** (GCP, Azure) and implemented scalable, cost-efficient data solutions, demonstrating strong knowledge in cloud-based infrastructure and its business applications.
- Led the implementation of **Metabase** as an open-source BI tool, streamlining data accessibility and reducing reliance on costly software solutions.

Professional Skills

Prog/Dev	Python, SQL, R, MATLAB, Java, C#, Git, Docker, Kubernetes, CI/CD, Linux, LaTeX
DE/Cloud	ETL Pipelines, Data Modeling, Airflow, Kafka, APIs (Salesforce, REST), GCP (BigQuery, Cloud Functions, Cloud Storage), Azure
DB/Vis	BigQuery, PostgreSQL, MS SQL Server, MySQL, Apache Cassandra, Tableau, Power BI, Looker, Metabase, Streamlit
ML/AI	PyTorch, TensorFlow, Predictive Modeling, Data Analysis, Statistical Modeling, NLP (GPT, Gemini APIs), LLM Integration
Soft Skills	Communication, Team Collaboration, Problem Solving, Agile Methodologies, Stakeholder Management, Leadership

Education

3.8 / 4.0	M.Sc. in Computer Science , Georgia Institute of Technology Atlanta GA (Remote)	2025
3.9 / 4.0	M.Eng. in Chemical Engineering , University of Calgary Calgary AB	2020
3.8 / 4.0	B.Eng. in Chemical Engineering , University of Calgary / Southwest Petroleum University (Cohort) Calgary AB	2018

Recent Projects

Natural Language Question Responding Model Dec 2024

- Natural Language Processing Georgia Institute of Technology
- Built and trained a **Key-Value Memory Network** using **PyTorch** to process structured data and generate accurate responses to natural language questions.
 - Demonstrated the ability to apply **Large Language Models (LLMs)** for generating insights, aligning with Amazon's focus on leveraging AI tools for business intelligence.

American Sign Language Recognition with Hidden Markov Models Jan 2024 - May 2024

- Artificial Intelligence Georgia Institute of Technology
- Designed and implemented an **AI agent** leveraging **Hidden Markov Models (HMMs)** to recognize American Sign Language (ASL) gestures from video input.
 - Employed **data preprocessing** and statistical modeling to deliver accurate gesture recognition, showcasing experience in processing complex datasets for actionable outcomes.

Doctoral Pathways for Online Master's Graduates May 2025

- Research & Data Analysis Georgia Institute of Technology
- Analyzed trends in PhD enrollment among online Master's graduates using **statistical analysis** and **educational data**.
 - Generated actionable insights and presented findings at ACM CompEd '25, demonstrating expertise in **data-driven decision making** and stakeholder reporting.